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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 05/03/2011 |
| **PI Name:** | Minna, John D. M.D. | | |
| **Project Title:** | NSCOR: Lung Cancer Pathogenesis and HZE Particle Exposure | | |
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| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
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| **Zip Code:** | 75390-8593 | **Congressional District:** | 30 |
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| **COI Name (Institution):** | Burma, Sandeep ( The University of Texas Southwestern Medical Center )<br>Chen, David ( The University of Texas Southwestern Medical Center )<br>Gazdar, Adi ( The University of Texas Southwestern Medical Center )<br>Girard, Luc ( The University of Texas Southwestern Medical Center )<br>Richardson, James ( The University of Texas Southwestern Medical Center )<br>Shay, Jerry ( The University of Texas Southwestern Medical Center )<br>Story, Michael ( The University of Texas Southwestern Medical Center )<br>Wright, Woody ( The University of Texas Southwestern Medical Center )<br>Xie, Xian-Jin ( The University of Texas Southwestern Medical Center )<br>Minna, John ( The University of Texas Southwestern Medical Center )<br>Yang, Chin-Reng ( The University of Texas Southwestern Medical Center ) | | |
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| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They also measure effects on development of cancer stem-like cells (cancer initiating cells). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified and that specific mRNA expression profiles for HZE irradiation exist. In addition, we have found that transgenic mouse models of lung cancer have increased lung cancer risk particularly after fractionated high LET exposure allowing us to study the early onset and progression of lung cancer in the whole animal in real time.<br>The NSCOR has 4 Projects:<br>1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation;<br>2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells;<br>3. Effect of HZE particle irradiation on functional progression of human lung cancer at the cellular and organotypic level;<br>4. Effects of HZE particles on the development of lung cancer in vivo in novel mouse models.<br>These projects are supported by 4 Cores: Administrative; Cell Culture; Genomics and Proteomics; and Biostatistics and Bioinformatics. In addition, with funding from the Department of Energy (DOE) contributing to this NSCOR, studies of low-LET and low dose radiation in the same model systems are also being undertaken. The HZE particle irradiation is done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, and a Center for the NCI Early Detection Research Network (EDRN) program, providing great synergism and additional resources available for the successful completion of this proposal. | | |
| **Rationale for HRP Directed Research:** | | | |

**Research Impact/Earth Benefits:**

Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in life time never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. In addition, in the past decade in the USA, Europe, and Asia there has been a resurgent interest in radiographic screening for lung cancer, particularly using serial computed tomography (CT) scans. In fact, the USA NCI has completed a large (50,000 person) National Lung Cancer Screening Trial (NLST) that randomized persons to have CT scans or not for the early detection of lung cancer. This trial has found a significant survival benefit for persons receiving CT scan screening. This raises a variety of questions concerning the use of this approach because of its cost and substantial population based radiation exposure. Greater than 25% of persons screened (current or former smokers) will have some lesions requiring further CT scan follow-up, when biopsies are performed only 1% of the overall group will have histologic evidence of lung cancer. In addition, many of the persons will require several CT scans in follow-up to determine if the lesion is growing to warrant a biopsy. Thus, there is considerable interest in knowing the contribution of CT scan doses of gamma radiation to the pathogenesis of lung cancer. This is particularly true since this radiation is being administered to individuals (current and former smokers) who are at highest risk of having preneoplastic lesions leading to lung cancer. Our studies of low dose radiation of the preclinical human and mouse model systems will allow us to provide some risk assessment for the effect of such radiation doses (low dose) on the genesis of lung cancer and provides the rationale for DOE support of our studies. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 50 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as expression of oncogenic K-RASv12, knockdown of expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high-LET and low dose gamma irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung cancer as part of long-term missions in space. In addition, as stated we will be developing risk estimates following low dose gamma radiation as seen on Earth, particularly from CT scan studies. Thus, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through CT scan screening studies, and environmental exposures, accidentally or through terrorism. A part of this will be to understand the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional (3D) cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes. Our large HBEC panel coupled with new high density single nucleotide polymorphism (SNP) analysis will allow describing polymorphic differences in such responses. Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions: "We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models ; "We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate? ; "We are developing tissue specific risk models using human 3D culture and animal models for lung cancer ; "We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, altered senescence, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated.
Overall we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. As will be described in this report, we are confident and will provide compelling evidence that the sum of the efforts of our assembled team are significantly contributing more to assessing cancer risk associated with space radiation compared to the individual components.

**Task Progress:**

Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation. Michael Story, Ph.D. Lead; Adi F. Gazdar, M.D. Co-Lead; John D. Minna, M.D. Co-Lead.
Specific Aim 1: To quantitate the effects of HZE particle irradiation in causing genetic changes in HBECs and their isogenic derivatives.

Specific Aim 2. To quantitate the amount of epigenetic changes caused by HZE particle irradiation in HBECs and their isogenic derivatives using high throughput quantitative methylation specific including genome wide and methylation specific PCR (MSP) assays for acquired promoter methylation.

Specific Aim 3. To quantitate the effects of HZE particle irradiation in causing expression profile (mRNA and protein) changes in human bronchial epithelial cells (HBECs) and their isogenic derivatives containing various lung cancer-related mutations using array based genome wide and multi protein detecting approaches.

Specific Aim 4. To determine the effect of inter-individual variation on the effects of HZE particle irradiation in causing genetic changes, epigenetic changes, mRNA and protein expression profile changes in human bronchial epithelial cells (HBECs).

A major goal of this project was to determine the radiosensitivity of normal HBECS to particles of increasing Z and energy in order to determine the LET that elicits the maximum biological response. The RBEs for fraction of cell survival was determined at a value of 0.38, and to date we find the maximal RBE is at or near 150 keV/u. As a result, other combinations of Z and energy are necessary to determine where the maximal LET is actually achieved. These experiments are ongoing.

The question of particle quality is being determined using cell survival as the anchor for biological response. Gene expression at biologically equivalent dose and at equal physical dose is being examined as a way to identify the mechanism for initial cellular response as well as to identify signaling pathways that may lead us to understand subsequent cell fate associated with the oncogenic process.

When whole genome analysis of gene expression was performed to compare gamma-ray, 1 GeV Fe and 1 GeV Si particles, we found a very common molecular response to such exposures. This response was quite evident when examining the p53 response following radiation and the subsequent signaling pathways that are initiated by increases in p53 transcription. Over a 24 h time period the responses of 4 genes downstream of p53 are very similar. Also, for each radiation species there are also distinct sets of genes that respond uniquely. As we expanded into other ions, O at 120 MeV and 1000 MeV in particular, to begin examining beam quality, we found that O at 120 MeV clustered with Si at 1000 MeV each with LETs of about 40 keV/u, while O at 1000 MeV (14 keV/u) segregated into its own cluster. We are still trying to determine if LET drives the molecular response.

Project 2: Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells. David Chen, Ph.D. Project Lead; Aroumougame Asaithamby, Ph.D. Co-Lead.

Aim 1. To test the hypothesis that the immediate cellular response to DNA damage induced by HZE would be different and more complex than that induced by X-rays. Aim 2. To test the hypothesis that persistent DNA damage induced by HZE is more complex and difficult to repair than that induced by X-rays and, therefore, more deleterious.

Aim 3. To determine if damage induced by HZE is preferentially repaired by non homologous end joining (NHEJ) or homologous recombination (HR) or is preferentially repaired in certain phases of the cell cycle.

Aim 4. To compare the response of human bronchial epithelial cells (HBECs) as non immortalized, immortalized, and with premalignant oncogenic changes in 3D cultures to HZE exposure using a high throughput approach.

Summary of Current Progress:

1. A majority of the complex DNA DSBs induced by iron-particles are irreparable in organotypic 3D cultures. To investigate the cellular responses to DNA DSBs induced by low- and high-LET IR in vivo, we generated a human bronchial epithelial cell (HBEC3) line that stably expresses near-physiological levels of EGFP-53BP1 and used this human lung epithelial cell line in a three-dimensional (3D) model tissue culture system (Matrigel) to study induction and repair of DNA DSBs under conditions that were close to those in 3D tissues in vivo by monitoring the formation of 53BP1 foci development and resolution with by monitoring the EGFP labeled protein. We found a majority of the complex DNA DSBs induced by Fe particles were irreparable in organotypic culture as compared with monolayer cultures. Further, most of the cells containing these irreparable DNA lesions are differentiated.

2. Irreparable complex DNA DSBs results in the generation of chromosome aberrations: To investigate how impaired DSB repair may lead to the initiation of carcinogenesis, we evaluated chromosomal aberrations in metaphase cells derived from the monolayer and organotypic cultures. Classical chromosome analysis of metaphase spreads revealed that at equal survival dose, the levels of chromosomal aberrations per mitotic cell was higher in both gamma-ray and Fe particle treated 3D structures relative to 2D culture. Further, the extent of chromosome aberrations was higher in cells derived from the organotypic culture as compared with monolayer culture.

Project 3. Effect of HZE Particle Irradiation on Functional Progression of Human Lung Cancer. John D. Minna, M.D. Lead; Michael Story, Ph.D. Co-Lead; Jerry Shay, Ph.D., Co-Lead

Specific Aim 1. Effects of HZE particles on early events in lung cancer progression (soft agar colony formation and other markers of cellular transformation).

Specific Aim 2. Effects of HZE particles on intermediate events in lung cancer progression (invasion in 3 D organotypic cultures).

Specific Aim 3. Effects of HZE particles on late events in lung cancer progression (ability to form tumors in immune deprived mice).

We have made substantial progress in the determination of transformation by HZE particles in normal human bronchial epithelial cells (HBEC3KT). (Transformation is defined by the ability to grow in soft agar.) We are examining oncogenic potential using cellular transformation as a surrogate by irradiating cells as a function of particle quality. To date we have compared cellular transformation with 1 GeV Fe and Si with gamma-rays.

Experiments with other particles are ongoing. We have determined that the maximum expression of cellular transformation is at or near 0.25 Gy when HBEC cells from two individuals were irradiated with 1 GeV Fe particles. The RBE (ratio of slopes between Fe and gamma-ray) is approximately 40, which is in line with tumorigenesis studies from heavy particles. We are in the process of isolating soft agar colonies and characterizing them for growth, ability to recapitulate the phenotype of growth in soft agar, performing transcriptional and chromosomal analysis, and testing them for tumorigenicity in NOD/SCID mice (experiments ongoing).

To date 196 transformed cell clones have been isolated, placed into short term cell culture, and then re-tested for growth in soft agar. Approximately half recapitulate this phenotype. Very distinct morphologic differences in some of the transformed cells isolated have been identified and which are distinguished by the genetic background of the originally irradiated cell. For instance, HBEC3KT cells, show the typical cobblestone appearance of epithelial cells grown in vitro. This is irrespective of whether they were irradiated or not. However, as seen in the lower panels of the figure, the unirradiated cells express the protein E-cadherin, which is associated with epithelial cells, but they do not express vimentin, which is associated with mesenchymal cells. This is contrasted by the irradiated cells, center and right panels, that express E-cadherin but also express vimentin. These irradiated cells are likely initiating epithelial to mesenchymal transition (EMT).

When transformed clones from irradiated HBEC3KTR53 cells (RAS mutant expressing, p53 down-regulated) where examined, 3 of the 4 transformed clones isolated exhibit a distinct morphology of narrow, elongated and overlapping cells that is associated with a mesenchymal phenotype (epithelial to mesenchymal transition, EMT), and furthermore, they do not express E-cadherin but do express vimentin, another hallmark of mesenchymal cells. These clones have clearly undergone EMT and this may be related to their RAS status.

We examined the irradiated HBEC3KT clones for RAS mutations and found none, although we did identify the MDM2 and CDK6 proteins as upregulated, suggesting enhanced proliferation capacity. We are examining p53 protein expression and will examine the p53 gene for mutations.

Project 4. Effects of HZE Particles on the Development of Lung Cancer in Novel Mouse Models. Jerry W. Shay, Ph.D. Project Lead; James Richardson, DVM, Ph.D. Co-Investigator; Oliver Delgado, Student Assistant

Specific Aim 1. Use the K-RAS transgenic mouse model of human lung cancer to determine effects of low- and high-LET radiation in tumor initiation and progression.

Specific Aim 2. Use additional transgenic mouse models to determine the effects of low- and high-LET radiation in both living animals and in isolated tissues.

The primary objective of using this mouse model of lung cancer susceptibility is to determine if there is an increased frequency of lung cancer initiation or progression after exposure to low- or high-LET radiation. Both mutant and wild-type animals have been exposed to a total dose of 1.0-3.0 Gy x-ray and 0.2 to 1.0 Gy 56Fe-. Experimental groups were further divided by the dose rate administered with one group receiving 1.0 Gy in one day and another group receiving 0.2 Gy per day for five sequential days. In the most recent NSRL run, we completed another cohort receiving 0.1 Gy 56Fe- per day for 5 days followed by a two day recovery and then an additional 0.1 Gy 56Fe- for 5 days. These animals received a total of 1 Gy of 56Fe- but fractionated in 10 doses over 12 days. We also irradiated another group of mice with single dose 0.2 Gy 56Fe-. These animals are being compared to an unirradiated cohort of wild-type and mutant animals. This animal model may help provide specific answers about risk assessment to questions outlined in the Bioastronautics Critical Path Roadmap (BCPR). Observations obtained post-irradiation of these mice may help determine the probability for increased pulmonary carcinogenesis from space radiation as a function of NASA operational parameters (age at exposure, radiation quality, dose rate, gender). Correlating the observed interactions with the DNA damage response and possible persistence of this damage compared to our 3D organotypic models. These experiments may help identify the molecular, genetic, epigenetic, or other factors contributing to lung carcinogenic risk.

Results from each of the above experiments are steadily emerging as these animals age and tissues are assessed. Even though the experiments are ongoing several notable observations are emerging. Irradiation of wild type littermates with a fractionated dose but not a single dose of 1.0 Gy 56Fe particles significantly decreases their survival compared to unirradiated wild-type animals. Expression of oncogenic K-RAS appears to eliminate any dependence on the method of radiation administration as mutant animals irradiated with either a fractionated or single dose of 1.0 Gy 56Fe particles have significantly decreased lifespan compared to unirradiated mutant LA1 K-RAS mice. In comparison, irradiation with 1.0 Gy x-rays does not affect the survival of mutant or wild type littermate LA1 K-RAS mice regardless of administration method. However, irradiation with fractionated 3.0 Gy x-ray does affect survival of both mutant and wild type littermates.

Examination of lung tissue extracted at death reveals that irradiation with any type of radiation does not appear to affect overall tumor incidence in irradiated mutant LA1 K-ras mice compared to unirradiated mutant

controls. Histopathological analysis of tumors from mutant LA1 K-ras mice irradiated with a fractionated dose of 56Fe particles, however, demonstrates a dramatic progression in the grade of lesions compared to unirradiated mutant mice, X-ray irradiated mutant mice, or mutants irradiated with a single 1.0 Gy dose of 56Fe particles. While K-ras mice that received a fractionated 1.0 Gy dose of 56Fe particles contain hyperplastic regions and adenomas similar to other experimental groups, the number adenocarcinomas are significantly increased. Subsets of these adenocarcinomas display highly aggressive and invasive characteristics of malignancy, which more closely mimics clinical features of lung cancer in humans. We are currently examining other tissues for metastasis in these mice.

In order to ascertain whether expression of oncogenic K-ras induces a differential pulmonary transcriptional response to irradiation, total RNA was extracted from lungs, which were flash frozen in liquid nitrogen three hours post- irradiation. Whole genome analysis was performed utilizing Sentrix(R) Mouse-6 Expression BeadChips (Illumina) by the NSCOR microarray Core, under the direction of NSCOR Investigator Dr. Michael Story. Unsupervised cluster analysis of resultant gene expression profiles has suggested divergent responses between mutant and wild type littermate lung associated with radiation type. Hierarchal clustering reveals unique genetic responses to radiation schedules such that fractionated 1Gy HZE irradiation was distinct from that of unirradiated mice and acute 1 Gy irradiated mice. These distinct signatures were obtained from whole lung RNA 70 days after irradiation. This eliminated any acute effects of IR and is approximately 200 prior to the histological observation of invasive cancer. Principal component analysis (PCA) suggested that the microarrays signatures correlated with radiation-induced progression. We next extracted the expression values from annotated human mRNA datasets, built classification trees using RPART and then used these gene classifier to predict high risk and low risk groups from these data sets and then plotted these as Kaplan-Meier survival graphs. The results demonstrated that fractionated but not acute dose classifiers predicted lung cancer survival in three independent microarray data sets. This is the first evidence that signatures of radiation induced cancer in mice can be used to model human cancer prognostics.

DOE Related studies: One aspects of our NSCOR is supported by the DOE. We are examining the effects on cellular transformation and on mRNA expression patterns of HBECs from low dose gamma irradiation.

Core A: Administrative Core. Leads: John Minna and Jerry Shay, and UTSW NSCOR Administrator: Brenda Zielke. The Administrative Core has established accounts and control over disbursement of all funds from this NSCOR application. It has also kept track of all personnel employed by this NSCOR. It has provided a central place for communication and coordinated data sharing In addition, it has established a series of biweekly to monthly scientific and planning meeting of the NSCOR investigators at UTSW. Approximately 25 scientific personnel attend each of these meetings. At each meeting administrative matters are discussed, experiments are planned, administrative changes made by the senior leadership team, and most importantly a scientific presentation by one of the NSCOR investigators is made and critiqued by the other investigators. Dr. Minna serves as the Administrative Core Lead with Dr. Jerry Shay as the Administrative Core Co-Lead. The Administrative Core coordinated all of the arrangements for the participation of several NSCOR investigators for a series of NSCOR mandated meetings in 2010, and the Administrative Core coordinated travel of multiple personnel to BNL for experimental work.

Core B. Cell Culture. Core Leads: John D. Minna, M.D. and Jerry Shay, Ph.D. Enhancement of human bronchial epithelial cell 3D lung model of tissue like development. There is increasing evidence that cancer arises from stem cell populations that are important for the normal turnover of tissues and the rapid repair of tissues in instances of injury. In the lung there is some evidence for a bronchiole alveolar stem cell (BASC) population that may give rise to lung carcinoma. We developed a panel of immortalized human bronchial epithelial cells and have previously demonstrated that these cells when placed in an air medium interface overlying a fibroblast stromal collagen matrix could differentiate into mucous and ciliated bronchial epithelial cells. These results were expected since the cells originated from the central airways and were obtained from midsized bronchial tissues. We examined these cells for a variety of differentiated cell markers and observed that they expressed p63 (a transit amplifying cell marker) and surprisingly surfactant protein A (SP-A), CC10, a Clara cell marker, and when differentiated weak expression of the thyroid transcription factor TITF1 (known to be important in adenocarcinomas of the peripheral airway of the lungs). These peripheral markers associated with Clara and type II alveolar cells imply that our immortalized HBECs have enhanced plasticity and should be considered pluripotent lung stem cells. We next placed these cells into Matrigel overlying a lung fibroblast (IMR90) feeder layer and observed that the cells could efficiently differentiate into spheroids or cyst like structures. Studies of these HBECs growing in matrigel and forming cysts and SACs indicated they started to express SP-A and morphologically resembled small bronchioles and alveolar sacs. When examined in the electron microscope, these cells also had lamellar bodies which are a hallmark of type II alveolar cells. This new improved 3D model of lung cell differentiation has many advantages over the previous model. First, they can differentiate in 3-5 days in vitro making it feasible to set these up at BNL for future NSRL runs. This is in contrast to the air liquid interface for differentiation into ciliated and mucous secreting bronchial cells which require a minimum of two weeks of culture and our previous experience indicated that they do not ship well to BNL. Second, these differentiating small airways cells resemble the type of lung cancer that is likely to be increased in astronauts on long-term space missions. Thus, during the 2008 and 2009 NSRL runs we initiated our irradiation experiments with these cells (see Project 2 overview). Importantly, we introduced a reporter gene 53BP1 containing EGFP so that they sites of irradiation induced damage could be visualized in real time. One final improvement in the 3D culture of our immortalized HBECs consists of culturing them on top of Matrigel. Under these conditions, the cells form web like structures then bud as if they are formed alveoli. This new method also permits a way to have 3D cultures in 5 days and thus promises to be a new addition to our arsenal of ways to examine the influence of irradiation on lung cells under various conditions.

Core C. Genomics Core. Core Lead: Michael Story, Ph.D. In collaboration with Core D: Biostatistics and Bioinformatics, has published a third article on background correction and data normalization for Illumina whole genome expression arrays1. This algorithm has been incorporated into BioConductor, the online bioinformatics package used throughout the bioinformatics community.

The core has also developed the technology to examine whole genome array-comparative genomic hybridization (aCGH) using the Nimblegen platform. The desire to examine copy number variation in mouse samples, project 4, was the reason for developing this platform. We can also overlay aCGH and expression data to quickly identify the differences in gene expression associated with copy number variation, and, Kaplan Meyer survival analysis techniques are available to identify chromosomal regions associated with survival analysis.

NSCOR Core D: Biostatistics and Bioinformatics. Core Lead: Xian-Jin Xie, Ph.D.; Core Co-Lead: Luc Girard, Ph.D. and Yang Xie, Ph.D.

The Biostatistics and Bioinformatics Core (Core D) provides comprehensive biostatistics and bioinformatics expertise to ensure the statistical integrity and to optimize data analysis of the University of Texas Southwestern (UTSW) NSCOR studies. In doing so, Core D assists investigators with testable hypothesis formulations and proper planning of experiments, manages experimental data generated by NSCOR projects, conducts and advises data analysis, and assists in interpreting final results. To play an integrated role and to serve all the NSCOR projects in a timely manner, Core D biostatisticians Drs. Xian-Jin Xie, Luc Girard, Yang Xie and have participated in all monthly project meetings, ensuring that proper biostatistics and data management considerations are adequately taken during all phases of planned experiments.

In conjunction with the Genomics (Expression Profiling and Proteomics) Core (Core C) and the Cell Culture Core (Core B), the Biostatistics and Bioinformatics Core (Core D) has the following specific aims:

Specific Aims:

- To provide database support and expertise for the collection, storage, integration, and retrieval of all sorts of data generated from NSCOR projects.

- To provide valid and optimal statistical design and to conduct proper analysis required to address the specific aims of each project.

- To review background and rationale, to assist in the design, evaluation, and analysis of new research proposal arising from the individual projects.

- To assist the other Cores and Projects in the proper analysis and interpretations of different types of bioassays, particularly mega-data such as those generated from gene expression arrays.

- To assist in manuscript preparation and to review scientific submissions.

- To assist the NSCOR team in building radiation cancer risk models.

As we progress on all the NSCOR projects, significant amount of new data have been generated since our 2009 annual progress report. As a result of Core D effort, a comprehensive data management system has been established that allows efficient access and sharing of all NSCOR data.

**Bibliography Type:** Description: (Last Updated: 11/20/2025)


| | |
|---|---|
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